ROOT CAUSE ANALYSIS

Why the Job Card & Production Planning module is not moving

Candor Foods ERP (CFERP) · production plans v2 and job cards v2 · data as of 6 Oct 2026, 12:36 IST

599	1,362	0	3	618
production plans created 23 May – 4 Sep 2026	job cards created 23 May – 22 Sep 2026	plans created in the last 30 days	job cards created in the last 30 days	sales orders received in the same 30 days

1. Executive summary

The job card and production planning module was used heavily from late May to August 2026: 599 plans and 1,362 job cards were created. It has since stopped. The last plan was created on **4 Sep** and the last job card on **22 Sep**. In the last 30 days the factory received 618 sales orders (3,491 order lines) and created no plans and only 3 job cards. The ERP itself is still in daily use: 35 of 54 user accounts logged in during the last 30 days, and the new Stock Take module recorded 4,172 postings in September. People have stopped using this module; they have not stopped using the ERP.

There is no single cause. The module got harder to use during July and August, which was also its busiest period, and in September nobody was clearly responsible for it any more. The five root causes, in order of impact:

#	Root cause	Severity
RC1	**Job cards could not be closed.** A 0.1% mass-balance gate, plus accounting bugs, left batches and multi-stage job cards stuck. Only an admin could release them.	Critical
RC2	**Planning has gone back to Excel.** The floor-wise plan of 24 Sep holds floor, person, capacity and dispatch date for 1,686 order lines. The ERP plan cannot hold these, and 416 of those lines have no SO in the ERP at all.	Critical
RC3	**The people who must act are not told, and often cannot see the card.** Job cards created by Approve get no floor, so floor users cannot see them. The first notification arrived on 23 Sep, and nothing tells anyone when a stage unlocks.	High
RC4	**Manual entry in the working tabs is fragile.** The four tabs used on the floor (Allocation, Raw Material scanning, Accounting, Box printing) are filled in by hand on purpose: the product changes form between stages, and the data will later feed automated models. But Allocation needs Stock Take access and a floor on the card, and a short Wi-Fi drop wipes unsaved entries.	Medium
RC5	**Unstable rollout and no owner.** The data model was rebuilt several times while in use. Development moved to Stock Take and Samples in September, migrations are still pending on the live database, and no adoption KPI is tracked.	High

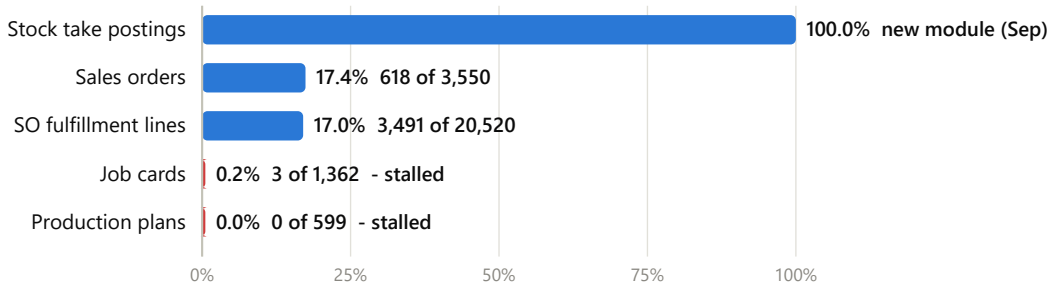
What to do first

- Clear the backlog of stuck job cards. The production head and floor managers go through every card older than 3 days and close, dispatch or cancel it, with a reason.
- Decide on one production plan, either the ERP or the Excel floor plan, and stop running both.
- Agree realistic weighing and process-loss tolerances for each product family with QC, and make them a written business rule.
- Appoint a module owner and review a weekly adoption KPI: % of SO lines planned, job cards closed, and job cards stuck for more than 3 days.

2. The problem in numbers

To separate "the module is idle" from "the factory is idle", the chart compares how much of each record type was created in the last 30 days. If use were steady over roughly four months, about 20–25% of all records would fall in the last 30 days. Sales orders are close to that (17%), so demand is steady. Plans and job cards are close to zero.

Share of all-time records created in the last 30 days (6 Sep – 6 Oct 2026)



Source: read-only queries on the production database (warehouse_db), 6 Oct 2026 12:36 IST. The figures include records later deleted or cancelled.

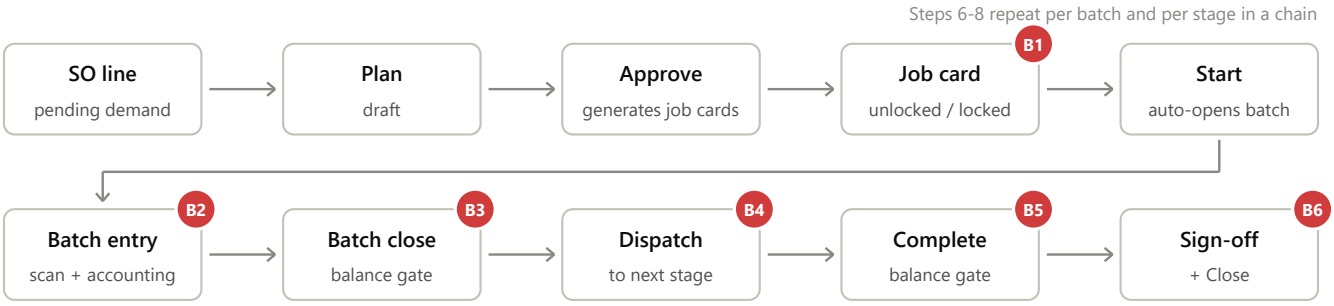
2.1 Timeline: how the module lost momentum

When	What happened	Type
22 May – 4 Jun	The last v1 plan (22 May) and the last v1 job card (4 Jun) are created. From 23 May all new work is on v2.	Change
4 Jun	Every Record output call returns HTTP 500 because a unique index is missing (migrations 016–029 were never run). Process steps are also inverted, with Packaging before Sorting.	Defect
15 Jul	Planning is merged into SO Creation and Plan List, and the separate Planning page is retired.	Change
18 – 31 Jul	Create-JC wizard, merging, partial carding, and a **batch balance lock** (30 Jul): a batch cannot close unless it balances to 0.1%.	Change
30 Jul – 20 Aug	**Double-count bug.** Dispatched output is counted twice, so every stage that dispatches becomes "structurally unclosable" and multi-batch job cards are stranded after batch 1.	Defect
21 – 26 Aug	The Output tab is removed. A second output save on the same batch returns 500 until 26 Aug. "Save completes the batch" arrives, and unbalanced batches now need an admin.	Defect
4 Sep	**Last production plan created.**	Stall
Sep	Development moves to Stock Take (8 commits, migrations 098–110) and Samples (17 commits). Only 2 job card commits.	Focus
22 Sep	**Last job card created.** JC BOM changes and floor requisitions ship, but need migrations 044, 112, 115 and 116 that are still pending on the live DB.	Stall
23 Sep	First notification: the floor manager is told when a job card is created. The WhatsApp template is still in review at Meta.	Change
24 Sep – 5 Oct	The "Pending orders – Floor wise plan" Excel (1,686 lines, 318 SOs) is the working plan. A draft migration (118) imports it independent of the SO and job card tables.	Shift

Dates before 2 Jul come from project documents; git history starts on 2 Jul (backend) and 7 Jul (frontend).

3. Where the workflow stalls

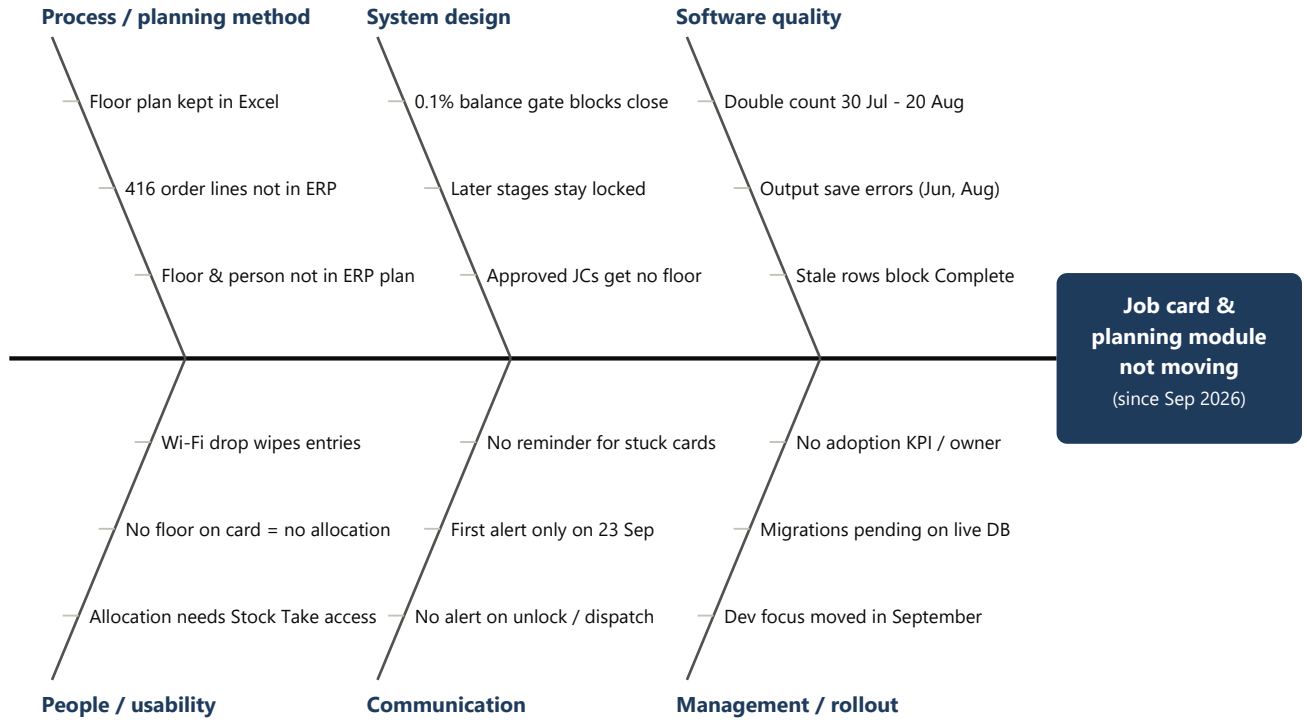
A job card goes through ten steps between a sales order line and closure. Steps 6–8 repeat for every batch and for every stage of a multi-stage chain (for example Sorting, then Roasting, then Packing). The red badges mark the six points where the code blocks people or slows them down.



	Step	What blocks or slows people	Evidence (file:line)
B1	Job card	Job cards created by Approve have no floor, so floor-scoped users cannot see them and no floor manager is notified. Later stages start locked and unlock only when the upstream stage dispatches, or by force-unlock. Cancelling the upstream card leaves them locked.	job_card_v2.py:974-978, 2837-2841, 4721-4723 · job_card_notify.py:300-306
B2	Batch entry	Four tabs are used on the floor: Allocation, Raw Material scanning, Accounting and Box printing. Entry is manual by design, because the product changes form at each stage and the data will later feed automated models. Allocation shows floor stock only to users with the Stock Take view permission, which is set up only for the stock_take role and admins, and does not work on a card with no floor. SAVE makes 4 calls in a row, and a short Wi-Fi drop replaces the card, and any unsaved form, with an error panel.	job-card/[id]/page.tsx:328-339, 5047-5285, 872-874, 965-971 · _MaterialAllocationTab.tsx:505-508 · 102_stock_take_rbac.sql:62-71
B3	Batch close	The batch must balance to within **0.1% of input**: 100 g on a 100 kg batch. A comment in the same code says ±50 g of eyeball weighing is normal. An unbalanced batch can only be released by an admin; there is no maker-checker route.	jc_accounting_v2.py:275-285 · job_card_batch_v2.py:338-360 · router.py:7152
B4	Dispatch	From 30 Jul to 20 Aug dispatched output was counted twice, so no dispatching stage could close. Rows saved in that window stay unbalanced until someone re-saves them. Over-dispatch errors come back as HTTP 200.	commit 2fe8cb8 · jc_accounting_v2.py:963-977 · router.py:6389-6393
B5	Complete	Requires no open shift, no open batch, and every accounting row balanced, including old card-level rows. One stale row blocks completion for good. The override needs an approved amendment (floor manager, then admin or production manager).	job_card_v2.py:4339-4409, 4476-4524 · router_amendments.py:53-82
B6	Sign-off / close	A production_head sign-off is required, but the production_head role has no permission grants. Anyone with sign-off rights types a name into a browser prompt. Close does not require Completed, so the balance gate can be bypassed. The rules are inconsistent.	job_card_v2.py:4890-4937, 4915 · 025_auth_positions.sql:22-25 · router.py:7474-7482

4. Root cause analysis

4.1 Cause-and-effect (fishbone)

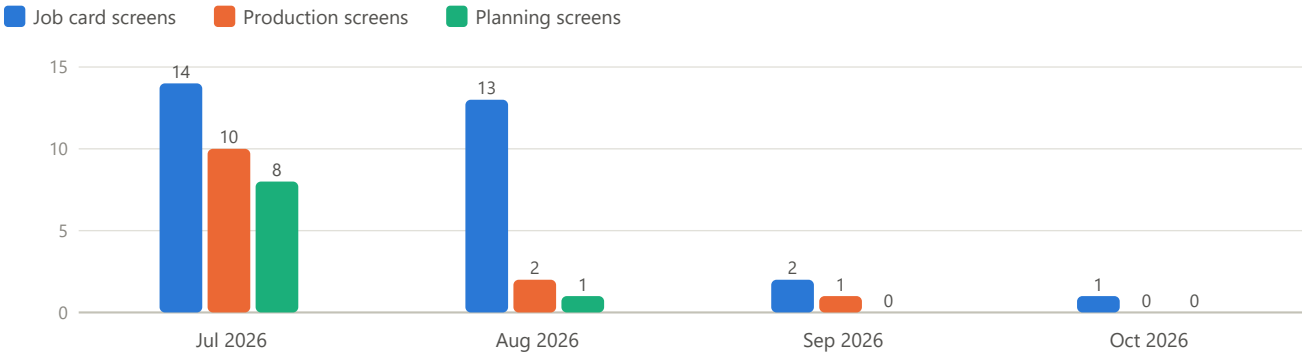


4.2 Five whys

Why?	Answer
Why are no job cards being created?	Because no plans have been created or approved in the ERP since 4 Sep. Job cards are generated only from a plan.
Why are no plans being created?	The planners' real plan is the Excel floor-wise plan. It carries floor, person, table capacity and dispatch date, which the ERP plan cannot hold. Keeping both plans is double work, so the ERP plan was dropped.
Why did the floor not keep the job cards going?	Job cards already on the floor got stuck. The balance gate and the double-count bug made batches and dispatching stages impossible to close without an admin (B3–B5). Later stages stayed locked. No one was told when a card was assigned or unlocked (B1).
Why was this not caught and fixed?	In September, development moved to Stock Take and Samples. Fixes were waiting on migrations not yet applied to the live database. Nobody tracked planned-vs-ordered or stuck job cards.
Root cause	**The module went live as a strict system of record before the floor process and master data were ready.** Floors were not settled and the real plan was still in Excel, and nobody was named as process owner to watch adoption and remove blockers.

4.3 Development effort on these screens

Frontend commits per month touching the module's screens



July includes the 7 Jul bulk import of earlier work. Backend job card commits: 14 in Jul, 23 in Aug, and 2 in Sep (22 and 23 Sep). Over the same September, Stock Take had 8 commits and Samples about 17.

4.4 Root causes in detail

RC1 · Job cards could not be closed

Critical

- The batch-close balance lock was added on 30 Jul with a 0.1% tolerance, and the code's own comment says operators eyeball-weigh to ±50 g. Unbalanced batches need an admin (jc_accounting_v2.py:275-285; job_card_batch_v2.py:338-360).
- From 30 Jul to 20 Aug the double-count bug made every dispatching stage unclosable. Example from the floor: JC 74337017 (Sorting) had 150.0 kg in, 149.8 out and a balance of −149.8 (commit 2fe8cb8). Unbalanced rows saved in that window remain until re-saved.
- Complete checks every accounting row, including stale card-level rows, so one bad row blocks a card permanently (job_card_v2.py:4395-4409). Before that: output saves returned 500 in June and again from 21 to 26 Aug.
  Effect: Floor teams learned that starting a job card in the ERP leads to a card they cannot finish. Stuck cards pile up and new ones are not started.

RC2 · Planning has gone back to Excel

Critical

- "Pending orders 24.09.2026 – Floor wise plan" assigns Packed to be, Executed By, Floor, Person and planned dispatch dates to 1,686 lines across 318 SOs. It also holds a capacity sheet (tables, manpower per table, output per shift).
- 416 of those lines (103 POs) have no SO in the ERP, so they cannot be planned there. The Excel floor names ("Upper, 1st Floor-M, Bar Line-H + F") do not match the ERP floor list.
- Draft migration 118 imports this Excel independent of so_header / so_fulfillment_v2. As drafted, it would make the parallel plan permanent rather than feed job cards.
  Effect: The ERP plan became a second copy of work already done in Excel, so people stopped making it, and without a plan no job cards are created.

RC3 · The people who must act are not told, and often cannot see the card

High

- Job cards created by Approve have a NULL floor. Floor-scoped users filter on floor, so these cards never show up for them (job_card_v2.py:2837-2841).
- There were no notifications at all until 23 Sep, and that one fires only on creation, via a WhatsApp template still in Meta review. Nothing fires on unlock, dispatch or pending sign-off, and there are no reminders.
- The idle-card and discrepancy checkers still read the v1 job_card table, so they never flag v2 cards (idle_checker.py:51-55; discrepancy_manager.py:31-37).
  Effect: Locked and unlocked cards wait silently. Nobody sees the backlog building up.

RC4 · Manual entry in the working tabs is fragile

Medium

- Four of the job card's tabs are used on the floor: Allocation, Raw Material scanning, Accounting and Box printing (job-card/[id]/page.tsx:328-339). They are filled in by hand on purpose: the product changes form between stages, and the data will later feed automated models. So the entry itself is not the problem; losing it or being blocked from it is.
- Allocation shows floor stock only to users with the Stock Take view permission, which is set up only for the stock_take role and admins. It does not work at all on a card with no floor, which is every card created by Approve (RC3) (_MaterialAllocationTab.tsx:505-508; 102_stock_take_rbac.sql:62-71).
- SAVE in Accounting makes 4 calls in a row. There is no offline support, and a short Wi-Fi drop replaces the card, and any unsaved entries, with an error panel (job-card/[id]/page.tsx:872-874, 965-971).
  Effect: When entries are lost or Allocation is blocked, floor staff fall back to paper, and the data meant for later automated models is never captured.

RC5 · Unstable rollout, no owner, no adoption tracking

High

- The data model changed repeatedly while in use: v1 to v2, the SFG vertical, the batch rework, accounting CRUD, and removal of the Output tab. v1 and v2 stacks still coexist, and the job card spec still documents v1 endpoints.
- Frontend commits to the job card, production and planning screens fell from 32 in Jul to 16 in Aug, 3 in Sep and 1 in Oct.
- Migrations 044, 112, 115, 116 and 117 are recorded as pending on the live database, and migrations 084, 085 and 087 were once skipped. The production_head role has no permissions, and floor managers cannot generate job cards.
  Effect: Problems found on the floor were not fixed or deployed quickly, and nobody tracked the module going quiet.

5. Who follows the process and who uses it

5.1 Who is expected to do each step (system roles)

Step	Permission	Roles that can do it
Create / edit / approve plan	plans.*	planner (and admin)
Generate job cards (wizard / merge)	job_cards.generate.create	planner, team_leader. **Not floor_manager**
Assign team / start / open batch	overview.assign, overview.start	floor_manager, team_leader
Record output & accounting	output.create, accounting.edit	floor_manager, team_leader
Close batch / complete / dispatch to next	overview.complete	floor_manager, team_leader
Dispatch group	job_cards.lifecycle.edit	team_leader only
Sign-off (as production_head)	sign_offs.create	floor_manager, team_leader, qc_inspector. **The production_head role itself has no grant**
Close job card	overview.close	floor_manager, team_leader
Overrides: cancel card, close partial, unbalanced batch, edit closed batch	admin only	admin

Source: 075_rbac_notes_catalog.sql:129-171, auth_schema.sql:244-297, router.py permission checks. Admins bypass every check (auth/middleware.py:160).

5.2 Actual usage by user (percentage frequency): pending

This section needs a read-only extract from the production database. That extract was not run for this draft. The query pack is ready (jc_usage_queries.sql, 18 read-only SELECTs, tested on the development database). Once its output is available this section fills in automatically with:

- each user's count of job card and plan actions, their % share of all actions, active days, first and last action, and what they did;
- adoption by role: active users, users who logged in during the last 30 days, and users who have ever used the module;
- distinct active users and actions per month, showing who stopped and when;
- job cards assigned per team leader and the % each completed;
- the job card and plan funnel, current status ageing by floor, and SO lines planned per month.

What is already known. All v2 plans were created between 23 May and 4 Sep 2026, and all v2 job cards between 23 May and 22 Sep. The ERP has 54 user accounts (49 active), and 35 of them logged in during the last 30 days. Only planners can create plans. Job cards can only be generated by planners or team leaders, not by floor managers. So the creation step depends on a small group of planners, and since 4 Sep none of them has created a plan.

6. Corrective and preventive actions

When	#	Action	Fixes	Owner (role)
Now (0–2 weeks)	1	Clear the stuck backlog. List the locked, unlocked and in-progress job cards older than 3 days. The production head and floor managers decide for each one whether to close, dispatch or cancel it, and record the reason.	RC1	Production head + floor managers
	2	Name one module owner for the whole sales-order → plan → job-card process, accountable for adoption and for removing blockers.	RC5	Plant head / management
	3	Decide on one production plan. Either the ERP plan or the Excel floor-wise plan is the official one, and both stop being kept in parallel. Until that is decided, write down which one the floors work from.	RC2	Plant head + planning head
	4	Agree weighing practice and acceptable process-loss tolerances for each product family with QC: which scale is used, how often, and what difference is acceptable. Record them as a written business rule.	RC1	Production head + QC
Next (2–6 weeks)	5	Plan only against orders that are booked. The 416 order lines in the Excel plan that have no SO are booked as SOs before they are planned.	RC2	Sales coordination / SO team
	6	Use one list of floor names across planning, Excel and the floors, and give every floor a named floor manager who is responsible for its job cards.	RC2 / RC3	Production head
	7	Write down who does what: who plans, who creates job cards, who signs off as production head, and who approves exceptions, with a named backup for each.	RC3 / RC5	Plant head + production head
	8	Hold a 5-minute open job card review at each shift handover on every floor, covering stuck and locked cards and pending sign-offs.	RC3	Floor managers
	9	Train planners and team leaders on the full flow, with one-page job aids. Explain why each manual entry matters (the product changes form, and the data will feed later automated models), and appoint one super-user per floor to help colleagues.	RC4	Production head + HR / training
	10	Collect structured feedback through a short walk-through with 2 planners and 2 team leaders per floor, to rank the pain points. This becomes the input for the separate ERP review.	RC4	Module owner
Sustain	11	Review the adoption KPIs every week in the production meeting: % of SO lines planned, plans approved, job cards closed, job cards stuck for more than 3 days, and active users per role.	RC5	Module owner / plant head
	12	Agree that every change to the floor process is announced and trained before it goes live, so the floors are never surprised.	RC5	Module owner

System (ERP) changes are not listed here. They will be analysed separately, using input from the team.

How we will know it worked

Measure	Today	Target (6 weeks)
SO lines planned in the ERP within 3 days of order	≈ 0% (no plans since 4 Sep)	≥ 80%
Job cards created per week	≈ 1 (3 in 30 days)	Matches planned SO lines
Job cards stuck more than 3 days (locked / unlocked / in progress)	Not tracked	< 10% of open cards
Batch closes needing an admin override	Admin only, not tracked	< 5% of batches
Floor managers and team leaders using the module each week	See Section 5	≥ 80% of those roles

7. Method, sources and limitations

Source	Used for
Production database (warehouse_db), read-only, 6 Oct 2026 12:36 IST	Headline counts: plans, job cards, sales orders, SO lines, stock take postings, user logins.
Backend code (FastAPI): production router and job card, batch, accounting and plan services; migrations	State machine, gates, tolerances, RBAC, notifications (B1–B8).
Frontend code (Next.js): job card, plan-list, SO Creation and planning screens	User journeys, data-entry burden, error handling, mobile fit.
Git history (backend from 2 Jul, frontend from 7 Jul 2026)	Timeline of changes and where development effort went.
Project documents: gap analysis (17 Apr), chain review (18 Apr), 500-error findings (4 Jun), convergence audit (21 Jun), Excel floor plan (24 Sep)	Earlier defects, whether they were fixed, and evidence of Excel planning.

Limitations. The causes come from the code, the change history, documents and aggregate counts. They have not been confirmed in interviews with planners or floor staff, and a short walk-through with two planners and two team leaders is recommended to rank them. User-wise percentages (Section 5.2) are pending the production extract. File and line references are to the working copy of 6 Oct 2026.